The Binomial Theorem

ClipSAT Printable Practice Suite (clipsat.org) • Topic 9.03

Part I — Free-Response Questions (FRQs): Foundational Fluency (1–7)

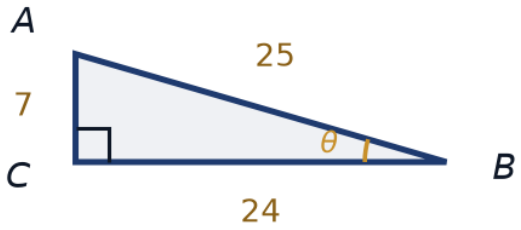
1 [Foundational] • Target: Binomial coefficients

Evaluate each binomial coefficient:

a $\binom{6}{2}$ **b** $\binom{8}{0}$ **c** $\binom{5}{5}$ **d** $\binom{10}{3}$

2 [Foundational] • Target: Binomial coefficients

Write down row 4 of Pascal's triangle (the row beginning $1, 4, \ldots$) and explain how it relates to the expansion of $(a + b)^4$.



3 [Foundational] • Target: Expansions and specific terms

Find the coefficient of x^3 in the expansion of $(2x - 1)^5$.

4 [Foundational] • Target: Expansions and specific terms

Use the binomial theorem to expand $(1 + 0.02)^5$ to the x^2-style second-order term, and hence estimate 1.02^5 to four decimal places.

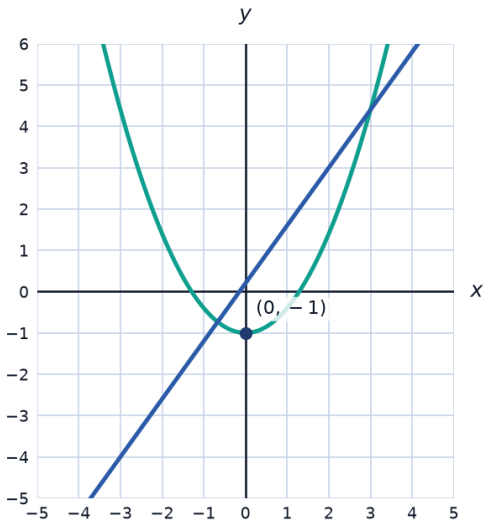
5 [Foundational] • Target: More expansions and terms

Find the middle term of the expansion of $(x + 3)^6$ (the term with x^3).

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6 [Foundational] • Target: More expansions and terms

Find the constant term in the expansion of $\left(x^2 + \frac{1}{x}\right)^6$.



7 [Foundational] • Target: Sums of binomial coefficients

Evaluate $\binom{5}{0} + \binom{5}{1} + \binom{5}{2} + \binom{5}{3} + \binom{5}{4} + \binom{5}{5}$ without computing each term, using the identity $\sum_{k=0}^{n} \binom{n}{k} = 2^n$.

Part II — Free-Response Questions (FRQs): Intermediate Multi-Step Application (8–14)

8 [Intermediate] • Target: Sums of binomial coefficients

Use the fact that $\binom{n}{k} = \binom{n}{n-k}$ (symmetry of Pascal's triangle) to explain why $\binom{9}{2} = \binom{9}{7}$ without computing either value directly.

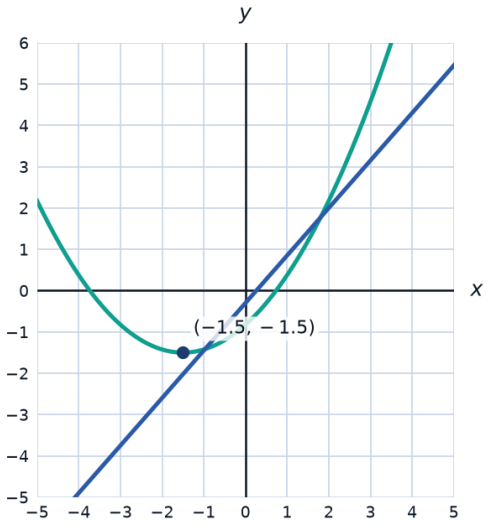
9 [Intermediate] • Target: Further expansions

Find the first three terms in the expansion of $(1 - 2x)^6$ in ascending powers of x.

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10 [Intermediate] • Target: Further expansions

Use the binomial theorem with $(1 - 0.01)^4$ to approximate 0.99^4 to four decimal places, using the first three terms.



11 [Intermediate] • Target: Deriving Pascal's triangle recursively

State Pascal's Rule, $\binom{n}{k} = \binom{n-1}{k-1} + \binom{n-1}{k}$, and use it to compute $\binom{6}{3}$ from row 5 of Pascal's triangle ($1, 5, 10, 10, 5, 1$).

12 [Intermediate] • Target: The general term formula

State the general term (the $(k+1)$th term) formula for the expansion of $(a+b)^n$, and use it to find the 5th term of $(x-2)^8$ directly (without expanding the whole thing).

13 [Intermediate] • Target: Combinatorial interpretation of binomial coefficients

Explain the combinatorial meaning of $\binom{n}{k}$, and use it to explain why $\binom{n}{0} = \binom{n}{n} = 1$ for any n.

14 [Intermediate] • Target: Applying the binomial theorem to probability-style counting

A fair coin is flipped 5 times. Using $\binom{5}{k}$, find the number of outcomes with exactly 3 heads, and state the total number of possible outcomes (using 2^5).

Part III — Free-Response Questions (FRQs): Challenging & Cross-Lesson Synthesis (15–20)

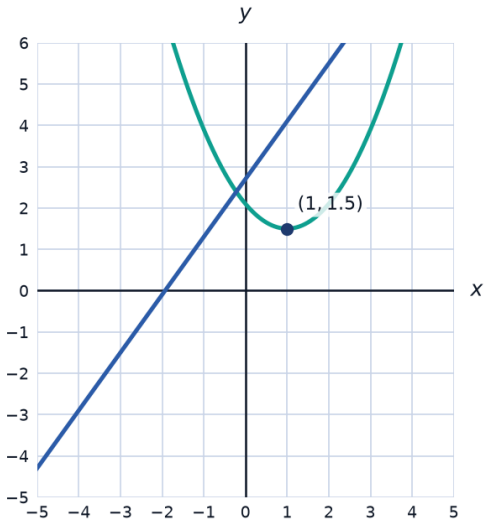
15 [Challenging] • Target: A closing challenge problem • Cross-Lesson: 5.01 Vectors: Components and Operations

Find the sum of all the coefficients in the expansion of $(3x-2)^4$, using the fact that setting $x = 1$ gives the sum of all coefficients.

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16 **[Challenging]** • **Target: A closing challenge problem** • **Cross-Lesson: 5.02 The Dot Product**

Find the coefficient of x^5 in the expansion of $(x^2 + 3)(x - 1)^6$, by finding the relevant terms from $(x - 1)^6$ first.



17 **[Challenging]** • **Target: A closing challenge problem** • **Cross-Lesson: 5.03 Parametric Equations**

Use the binomial theorem to expand $(x + h)^3 - x^3$ fully, then divide by h and simplify — this expression is exactly the difference quotient used to find the derivative of $f(x) = x^3$ in calculus.

18 **[Challenging]** • **Target: A closing challenge problem** • **Cross-Lesson: 6.01 Polar Coordinates and Conversions**

Evaluate $\binom{7}{2}$ and $\binom{7}{5}$, and confirm they are equal using the symmetry property.

19 **[Challenging]** • **Target: Applying the binomial theorem to probability-style counting** • **Cross-Lesson: 6.02 Polar Equations and Graphs**

A school committee of 4 members is to be selected from a club consisting of 6 juniors and 5 seniors. (a) In how many total ways can the 4 members be selected? (b) In how many ways can the committee be formed if it must contain exactly 2 juniors and 2 seniors?

20 **[Challenging]** • **Target: A closing challenge problem** • **Cross-Lesson: 7.01 Exponential Functions and Their Graphs**

A game pays $x15if a player rolls a 6 on a fair six-sided die, pays \6$ if they roll a 5, and the player must pay $x3$ for any other roll (1, 2, 3, or 4). Determine the expected gain or loss per roll, and state whether the game is favorable to the player.

Part IV — Multiple-Choice Questions (MCQs): Foundational Concept Mastery (21–28)

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21 [Foundational] • Target: Binomial coefficients

Three distinct vertices of a regular 12-gon are selected uniformly at random. What is the probability that the triangle formed by these three vertices is obtuse?

a. $\frac{3}{11}$

b. $\frac{5}{11}$

c. $\frac{6}{11}$

d. $\frac{7}{11}$

22 [Foundational] • Target: Expansions and specific terms

The first three terms of the binomial expansion of $(1 + kx)^n$ are $1 + 6x + 12x^2$. Find the value of k.

a. 2

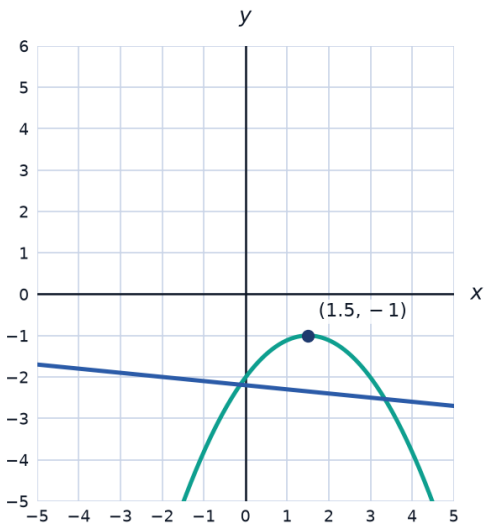
b. -2

c. 3

d. -3

23 [Foundational] • Target: More expansions and terms

A geometric series has first term $a = 11$ and common ratio $r = 1/2$. Find S_∞.



a. 11/2

b. 22

c. 22/3

d. 33

24 [Foundational] • Target: Sums of binomial coefficients

A fair coin is tossed 3 times. Let X be the number of heads obtained. What is the probability that X = 2?

a. $\frac{1}{8}$

b. $\frac{3}{8}$

c. $\frac{1}{2}$

d. $\frac{7}{8}$

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25 [Foundational] • Target: Further expansions

The bar graph below shows the distribution of customer ratings given to a local bakery by 20 customers on a scale from 1 to 5 stars. How many additional 5-star ratings must the bakery receive so that the overall mean rating of all customers becomes exactly 3.5 stars?

a. 6

b. 8

c. 12

d. 16

26 [Foundational] • Target: Deriving Pascal's triangle recursively

The bar graph below shows the number of students enrolled in each of four elective classes at a high school. Each student is enrolled in exactly one of these four classes. If one student is selected at random from these four classes, what is the probability that the student is enrolled in Drama?

a. $\frac{1}{4}$

b. $\frac{4}{25}$

c. $\frac{4}{21}$

d. $\frac{7}{25}$

27 [Foundational] • Target: The general term formula

A fair six-sided die is rolled 4 times. Let X be the number of times a 6 is obtained. The possible values for X are 0, 1, 2, 3, and 4. The probability of obtaining a 6 on any given roll is $p = \frac{1}{6}$. The number of trials is $n = 4$. This scenario can be modelled by a binomial distribution, denoted as $X \sim B(n, p)$. Which of the following represents the correct parameters for this binomial distribution?

a. $B(4, 1/6)$

b. $B(6, 1/4)$

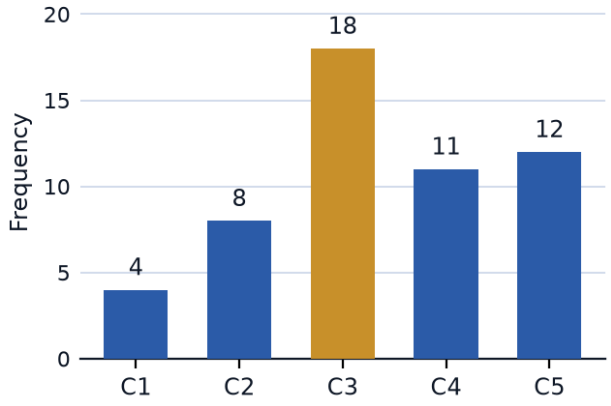
c. $B(1, 4/6)$

d. $B(4, 4/6)$

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28 [Foundational] • Target: Combinatorial interpretation of binomial coefficients

The bar chart below displays the number of cars sold by an auto dealership each day from Monday through Thursday. What was the mean number of cars sold per day over these 4 days?



a. 5.5

b. 6.0

c. 6.5

d. 8.0

Part V — Multiple-Choice Questions (MCQs): Intermediate Exam Application (29–37)

29 [Intermediate] • Target: Applying the binomial theorem to probability-style counting

State and prove the Bayes' theorem for continuous distributions.

a. Not valid for continuous distributions

b. Same as discrete case

c. $f(\theta|x) = \frac{f(x|\theta)\pi(\theta)}{\int f(x|\theta)\pi(\theta)d\theta}$ — from the definition of conditional density

d. Use the CLT

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30 [Intermediate] • Target: A closing challenge problem

Show that the binomial distribution approaches Poisson as $n \to \infty$, $p \to 0$, $np = \lambda$ fixed.

a. Requires $n > 100$

b. Not true

c. $\binom{n}{k} p^k (1-p)^{n-k} \to \frac{e^{-\lambda}\lambda^k}{k!}$ — use $(1-\lambda/n)^n \to e^{-\lambda}$

d. Only for $k = 0$

31 [Intermediate] • Target: Binomial coefficients

State and apply the Central Limit Theorem.

a. Only for discrete distributions

b. For large n, $\bar{X} \sim N(\mu, \sigma^2/n)$ approximately, regardless of the population distribution

c. Only for $n > 1000$

d. Only for normal populations

32 [Intermediate] • Target: Expansions and specific terms

A factory tests 30 independent components, each with a fixed probability of being defective; X is the number of defective components. Is X binomially distributed?

a. Yes — fixed, independent trials with constant success probability

b. No — the number of trials is not fixed

c. No — trials are not independent / probability is not constant

d. Yes — but only if $n > 30$

33 [Intermediate] • Target: More expansions and terms

State Bayes' theorem: $P(A|B) =$

a. $\frac{P(B|A)P(A)}{P(B)}$

b. $\frac{P(A)}{P(B)}$

c. $\frac{P(A)P(B)}{P(A|B)}$

d. $P(A)P(B|A)$

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34 [Intermediate] • Target: Sums of binomial coefficients

Of 27 students, 17 play soccer and 3 of those also play basketball. What is the probability that a randomly chosen soccer player also plays basketball?

a. $\frac{3}{10}$

b. $\frac{14}{17}$

c. $\frac{3}{17}$

d. $\frac{17}{27}$

35 [Intermediate] • Target: Further expansions

A machine's failure probability increases each hour it runs; X is the number of failures in a day. Is X binomially distributed?

a. No — trials are not independent / probability is not constant

b. Yes — fixed, independent trials with constant success probability

c. No — the number of trials is not fixed

d. Yes — but only if $n > 30$

36 [Intermediate] • Target: Deriving Pascal's triangle recursively

The Fundamental Theorem of Algebra states that every polynomial of degree $n \geq 1$ with complex coefficients:

a. Has n rational roots

b. Has at most n roots

c. Has real roots only

d. Has exactly n roots in $\mathbb{C}$ (counting multiplicity)

37 [Intermediate] • Target: The general term formula

Use De Moivre's theorem to express $\cos 4\theta$ in terms of $\cos\theta$.

a. $\cos^4\theta$

b. $4\cos^2\theta - 3$

c. $8\cos^4\theta - 8\cos^2\theta + 1$

d. $4\cos^4\theta - 3\cos^2\theta$

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Part VI — Multiple-Choice Questions (MCQs): Challenging & Cross-Lesson Synthesis (38–45)

38 **[Challenging]** • **Target: Combinatorial interpretation of binomial coefficients** • **Cross-Lesson: 6.01 Polar Coordinates and Conversions**

Derive the maximum likelihood estimator for p in Binomial distribution.

a. $\hat{p} = \sqrt{k/n}$

b. $\hat{p} = k/n$ where k is observed successes and n is total trials

c. $\hat{p} = 1$

d. $\hat{p} = n/k$

39 **[Challenging]** • **Target: Applying the binomial theorem to probability-style counting** • **Cross-Lesson: 6.02 Polar Equations and Graphs**

A bag has 13 marbles, 2 of which are red. What is the probability of drawing a red marble?

a. $\frac{11}{13}$

b. $\frac{2}{13}$

c. $\frac{1}{13}$

d. $\frac{2}{11}$

40 **[Challenging]** • **Target: A closing challenge problem** • **Cross-Lesson: 7.01 Exponential Functions and Their Graphs**

How many ways can you choose 5 items from 13 distinct items, where order does not matter?

a. 1286

b. 1288

c. 154440

d. 1287

41 **[Challenging]** • **Target: Binomial coefficients** • **Cross-Lesson: 7.02 Logarithms and Their Properties**

How many ways can you arrange 2 items chosen from 10 distinct items, where order matters?

a. 89

b. 90

c. 91

d. 45

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42 [Challenging] • Target: Expansions and specific terms • Cross-Lesson: 7.03 Exponential and Logarithmic Equations

The Poisson distribution arises as a limit of Binomial. For what parameters?

a. $n \to \infty$, $p = 1/2$

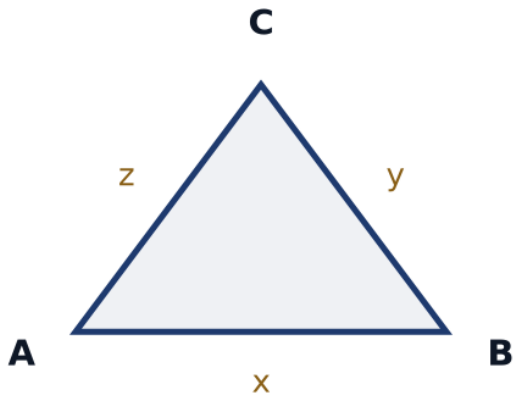
b. $n = 1$, any p

c. $p \to 1$

d. $n \to \infty$, $p \to 0$, $np = \lambda$ constant

43 [Challenging] • Target: More expansions and terms • Cross-Lesson: 7.04 Applications: Growth and Decay

The diagram shows a triangle ABC with coordinates A(2, 1), B(8, 1) and C(5, 5). Find the length of the side AC.



a. 5

b. x

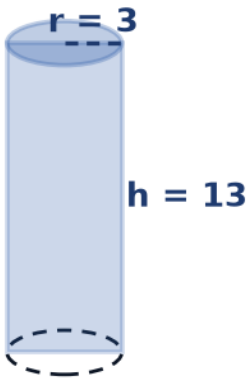
c. 6

d. 7

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44 [Challenging] • Target: Sums of binomial coefficients • Cross-Lesson: 8.01 Rational Functions and Asymptotes

A solid is formed by a cylinder of radius 3 and height 13 topped with a hemisphere of the same radius. Find the total volume. (In terms of π.)



a. 164π

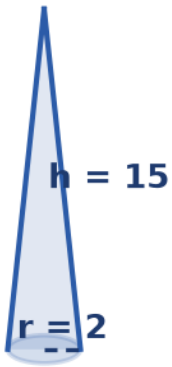
b. 27π

c. 135π

d. 51π

45 [Challenging] • Target: Further expansions • Cross-Lesson: 8.02 Graphing Rational Functions

Find the volume of a cone with radius 2 and height 15. (In terms of π.)



a. 4π

b. 2π

c. 32π

d. 20π

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